Bagging, Boosting and Measured Bias–Variance on Olivetti Faces

Sourabh Singh Roll no. M25CPS009
Indian Institute of Technology, Jodhpur

Machine Learning, Assignment 2

Code: https://github.com/Sourabh16/Machine-Learning-Assignment2-Bagging-Boosting

Abstract

I compare a single decision tree, bagging and multiclass SAMME boosting for face identification on raw Olivetti pixels. Bagging and boosting are my own implementations on top of scikit-learn's `DecisionTreeClassifier`*. On clean test faces the tree gets 55.0%, bagging (200 trees) 81.3% and boosting (depth 4, 100 rounds) 46.3%; with the centre of the face masked these drop to 23.8%, 67.5% and 35.0%. Over 30 resampled training sets bagging removes about 96% of the tree's variance, while boosting has the lowest variance but the highest bias and worst accuracy.*

1. Setup

Data and split. The seed is the digits of my roll number, `default_rng(25009)`. For each of the 40 people I shuffle the ten images and keep 6 for training, 2 for validation and 2 for test (240/80/80). The split is saved in `splits/` and reused by every script. I use the 4096 raw pixels with no PCA. The masked test set is a copy of the test images with rows and columns 24–39 set to zero (Fig. 1); training only ever uses clean images.

Implementation. Bagging and SAMME are in `src/ensembles.py`. Bagging draws a bootstrap of 240 images and a tie-breaking seed for each tree in turn, so the first B trees of the 200-tree model are exactly the B-tree model and every B can be read from one fit (refitting at 25/50/100 trees in Q2(c) gives the same numbers). It predicts by majority vote, and its confidence is the winning vote fraction.

Selection. Every choice uses clean validation accuracy only, with ties going to the simpler model. Test accuracy was only looked at after freezing.

2. Q1: Single tree and bagging

2.1. (a) Single-tree baseline

Table 1 shows the grid. For every capped depth the three leaf sizes give the same tree, since no leaf is small enough for the limit to matter. The best validation accuracy (53.8%) is for `max_depth=None`, `min_samples_leaf=4`, which I froze.

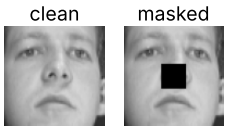


Figure 1. A test face and its masked copy (rows/cols 24–39 set to 0).

max_depth / leaf	train	val	test	leaves	per leaf	single
2 / any	7.5	7.5	7.5	3	80.0	0
4 / any	12.5	11.3	11.3	5	48.0	0
8 / any	22.5	18.8	21.3	9	26.7	0
16 / any	42.5	31.3	32.5	17	14.1	0
None / 1	100	48.8	55.0	46	5.2	0.13
None / 2	97.5	48.8	53.8	46	5.2	0
None / 4	**96.3**	**53.8**	**55.0**	**40**	**6.0**	**0**

Table 1. Single-tree grid (accuracy in %). "per leaf" is the mean number of training images per leaf and "single" the fraction of leaves holding exactly one training image. The frozen tree (bold) has realised depth 36.

What changes with depth. The trees are very lopsided: a tree capped at depth d has exactly $d + 1$ leaves (Fig. 2). Each split peels a few faces off and leaves one big mixed node, because one pixel threshold can separate one person from the rest but not 20 people from 20. So a depth-16 tree can name at most 17 of the 40 people, which is why training accuracy is only 42.5% and accuracy grows almost linearly with depth. Only unbounded trees (36–37 levels, 40–46 leaves) fit the data. With leaf size 1, 13% of leaves hold one image and training accuracy is 100%; leaf size 4 gives 40 leaves of about six images, roughly one per person, losing 3.8 training points but gaining 5 on validation. Deeper trees reduce bias, but at full depth they start memorising single images.

2.2. (b) Bagging from scratch

Table 2 gives the results. A bootstrap of size n should contain $n[1 - (1 - 1/n)^n] = 151.89$ distinct images ($0.633n$,

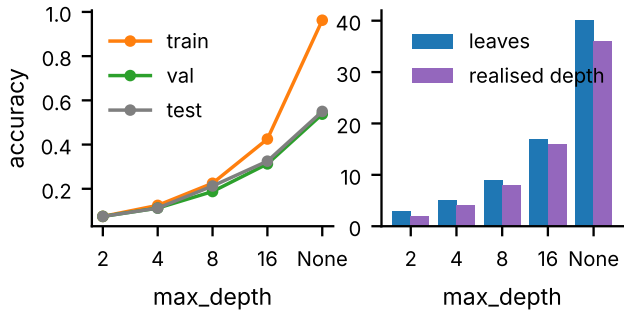


Figure 2. Left: accuracy against `max_depth` (`min_samples_leaf=4`). Right: leaves and realised depth. Leaves = depth+1 for every capped tree.

B	train	val	test	distinct	OOB	dis.	margin
1	67.9	40.0	40.0	154.0	31.4†	–	1.00
5	90.4	55.0	46.3	151.0	29.9†	.755	.270
10	98.8	72.5	58.8	153.4	39.8†	.778	.258
25	99.6	70.0	71.3	151.9	55.4	.799	.241
50	100	80.0	72.5	151.8	68.3	.790	.255
100	100	78.8	78.8	151.7	75.4	.793	.261
200	100	**81.3**	**81.3**	151.6	77.1	.801	.250

Table 2. Bagging the frozen tree. "distinct" is the mean number of distinct training images per bootstrap (theory 151.89). OOB, mean pairwise disagreement ("dis.") and mean vote margin use the first B trees; dis. and margin are measured on the validation set. †Only 36%, 89% and 98% of training images have at least one out-of-bag tree at $B = 1, 5, 10$.

close to $1 - e^{-1}$). Over the 200 bootstraps I measured 151.6 ± 4.7 (139–167), which matches. Training accuracy reaches 100% by $B = 50$, and test accuracy goes from 40.0% for one bagged tree (it sees only about 152 distinct images) to 81.3% at $B = 200$.

2.3. (c) OOB accuracy, disagreement and vote margin

Figure 3 shows the three quantities against B. Validation accuracy is highest at $B = 200$ (81.3%, against 80.0% at 50 and 78.8% at 100), so the final model uses 200 trees.

The three curves do not tell the same story. *OOB accuracy* tracks validation accuracy and is still rising at 200 (75.4%→77.1%). It sits a little lower because each image is out-of-bag for only about $0.37B$ trees, and below $B = 25$ it does not cover every image. *Disagreement* is flat at about 0.80 (two trees disagree on four of five faces), since it is a property of the base tree rather than of B. The *mean margin* falls from 1 to about 0.25 by $B = 5$ and then stays put.

So adding trees does not make each face more confidently classified; it makes the plurality more reliable, since the noise in which class wins shrinks like $1/\sqrt{B}$. Disagreement and margin describe the trees, which the tree setting fixes; OOB and validation accuracy show how far the averaging has converged.

	clean		masked	
	acc	conf	acc	conf
Tree (leaf probability)	55.0	.971	23.8	1.000
Bagging, B=200 (win. vote)	81.3	.356	67.5	.258
Bagging conf. if correct / wrong	.390 / .212		.301 / .167	
Bagging disagreement (test)	.810		.881	

Table 3. Frozen tree and final bagged model on the two test sets.

2.4. (d) Tree versus bagging, clean and masked

Table 3 compares the two. The tree is about as confident when wrong as when right (0.96 vs 0.98), so its confidence is useless. Bagging's winning vote is lower but meaningful (0.39 when right, 0.21 when wrong). Masking wrecks the tree: 54 of the 80 masked faces fall into a single leaf (identity 35), masked faces reach only 12 leaves instead of 37, and accuracy drops to 23.8% at confidence 1.0. Bagging loses 14 points, and its disagreement rises from 0.81 to 0.88, so it at least shows that it is less sure.

On clean faces bagging fixes 23 of the tree's mistakes, adds 2, and 13 are wrong for both (Fig. 4). **Fixed:** the tree calls face 338 (identity 33) "23" with confidence 1.0, but only 7 of 200 trees agree and 119 vote 33; face 175 (identity 17) is "3" for the tree but gets 112 votes for 17. These mistakes came from one particular path of thresholds, which other bootstrap trees do not repeat, so they average out: *variance* errors. **Both wrong:** face 234 (identity 23) gets 72 votes for 13 and only 6 for the truth, and face 251 (identity 25) gets 67 votes for 37 against 38. Here the trees agree on the wrong person, and the other image of identity 25 also goes to 37, so these two people simply look alike to the trees. Averaging cannot move the average prediction, so these *bias* errors remain.

3. Q2: Multiclass SAMME boosting

My SAMME uses $\alpha_t = \log \frac{1-\epsilon_t}{\epsilon_t} + \log(K-1)$, multiplies the weights of misclassified images by e^{α_t} and renormalises, predicts with $S_k(x) = \sum_t \alpha_t \mathbb{1}[h_t(x) = k]$ and uses $q_k = S_k / \sum_t \alpha_t$ as confidence. A learner with $\epsilon_t \geq 1 - 1/K$ would get $\alpha_t \leq 0$, so I reject it, keep the weights, and refit with a new seed, stopping after five rejections in a row. A perfect learner would get infinite α, so ϵ is clipped at 10^{-10} and boosting stops. Neither case occurred: after round 5 the largest error was 0.74, far below 0.975.

3.1. (a) One logged round on a toy problem

Table 4 is the round-one log from the same class used on the faces, checked against a hand calculation. A stump can name only two of the three classes, so all of class 2 is wrong, but $\epsilon = 1/3$ is still below the random level $2/3$, so $\alpha_1 > 0$.

One useful fact falls out of the update: the misclassified images always end up with exactly $(K-1)/K$ of the total weight, whatever ϵ_t is (here $2/3$). On the faces that is

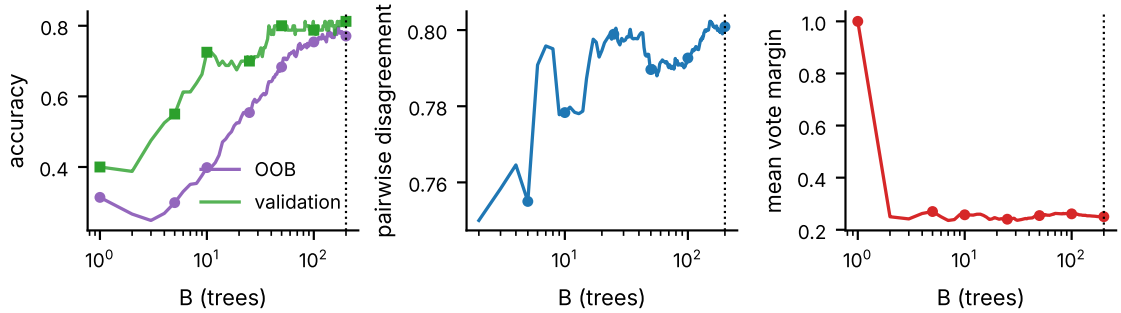


Figure 3. Bagging diagnostics against B (log scale). Lines are evaluated at every B, markers at the required sizes, and the dotted line is the selected $B = 200$. Disagreement and margin are on the validation set.



Figure 4. Top: test faces with the tree prediction and the top two bagging vote counts (out of 200). Bottom: a training image of the identity that was wrongly predicted (tree's guess for the corrected cases, bagging's for the failures).

i	x_i	y_i	$w_i^{(0)}$	$h_1(x_i)$	miss	$w_i^{(1)}$
0–2	(1,1),(1.5,2),(2,1.2)	0	1/9	0	0	0.0556
3–4	(4,4),(4.5,3.5)	1	1/9	1	0	0.0556
5	(2.2,1.6)	1	1/9	1	0	0.0556
6–7	(7,1),(6.5,2)	2	1/9	1	1	0.2222
8	(4.2,3.8)	2	1/9	1	1	0.2222

Table 4. One SAMME round on 9 points and 3 classes with a depth-1 stump ($x_1 \le 2.1 \Rightarrow 0$, else 1). $\epsilon_1 = 3/9$, $\alpha_1 = \log 2 + \log 2 = 1.386$; misclassified weights are multiplied by $e^{\alpha_1} = 4$ before renormalising.

$39/40$, so the images a learner got right keep only 2.5% of the weight between them.

3.2. (b) Depth and number of rounds

Table 5 shows the grid; depth 4 with 100 rounds is selected. Accuracy rises along both axes because the learners really are weak: a depth-d tree can name at most 2^d people (2 for a stump, 16 at depth 4), and stumps reach only 10%. At depth 4 the validation curve is still rising at 100 rounds, so the grid is the limit.

depth \ rounds	10	25	50	100
1	3.8	7.5	8.8	10.0
2	3.8	11.3	15.0	22.5
3	16.3	18.8	26.3	33.8
4	26.3	28.8	36.3	**50.0**

Table 5. Clean validation accuracy (%) of SAMME. The selected model (depth 4, 100 rounds) has 99.6% train and 46.3% test accuracy.

	M	train	val	test	time (s)
SAMME (depth 4)	25	60.8	28.8	30.0	5.7
	50	87.5	36.3	42.5	11.0
	100	99.6	50.0	46.3	22.5
Bagging (full trees)	25	99.6	70.0	71.3	17.5
	50	100	80.0	72.5	34.4
	100	100	78.8	78.8	69.5

Table 6. Equal numbers of learners, each model refitted from scratch on one core. Bagging's numbers match Tab. 2, which confirms the prefix property.

In Fig. 5, the weighted error starts at 0.85 and settles around 0.66: far below 0.975 but large, since a 16-leaf tree must miss at least 24 people. α stays near 3.0. Weights concentrate fast: the largest goes from 0.0042 to 0.054 (13 times uniform), and the ten heaviest images (4% of the data) hold 14–40% of the weight after round 20. The saw-tooth pattern is the $(K-1)/K$ rule: heavy faces that a learner gets right lose most of their weight at once, then regain it when later learners miss them.

3.3. (c) Boosting versus bagging at equal size

Table 6 compares them at equal size. Bagging is 25–40 points better everywhere, while boosting is about three times faster per learner (0.22 s vs 0.7 s per tree). Boosting reaches 99.6% on training data but only 50% on validation, with very small margins (0.026 against 0.23 for bagging).

A heavily weighted face. Training image 87 (identity 8) ends with the largest weight, 0.054 (Fig. 6). Only 12 of the 100 learners get it right; it is most often called 2, a similar-looking person. Each correct learner cuts its weight

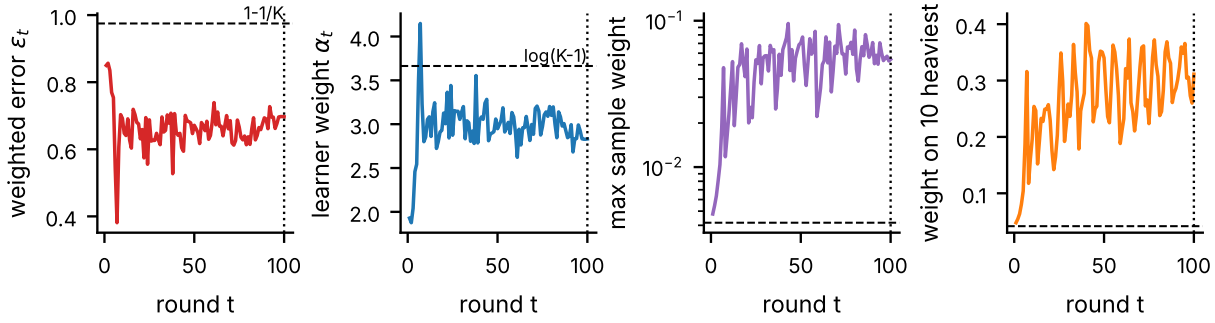


Figure 5. Selected SAMME run (depth 4): weighted training error (dashed: random level $1 - 1/K$), learner weight (dashed: $\log(K-1)$, *i.e.* $\epsilon = 0.5$), largest sample weight (log scale, dashed: uniform $1/240$) and total weight of the 10 heaviest training images (dashed: uniform 10/240).

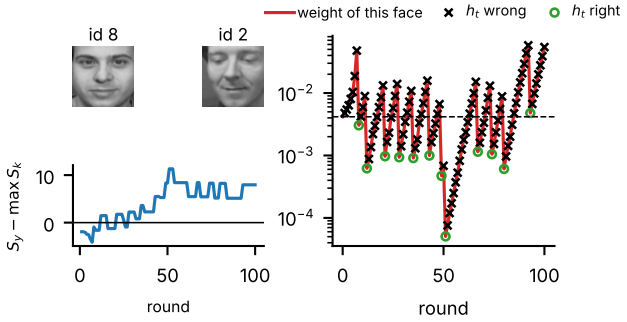


Figure 6. Training image 87 (identity 8) and a training image of identity 2, its most frequent confusion. Right: its sample weight per round (log scale) with the correctness of h_t. Bottom left: its ensemble margin $S_y - \max_{k \neq y} S_k$ over rounds.

	clean		masked	
	acc	conf	acc	conf
Single tree (leaf prob.)	55.0	.971	23.8	1.000
Bagging, 200 (win. vote)	**81.3**	.356	**67.5**	.258
SAMME, d4/100 (max q_k)	46.3	.114	35.0	.104

Table 7. Frozen models on clean and masked test faces. Confidence is the mean winning probability or vote. Accuracy drops: tree -31.3, bagging -13.8, boosting -11.3 points.

by more than ten times, but the next 16-leaf tree usually has no room for identity 8, gets it wrong, and multiplies its relative weight by $e^{\alpha} \approx 20$. So boosting keeps coming back to it, not because the label is wrong but because no single weak learner can hold it. The full ensemble does classify it correctly in the end.

3.4. (d) Robustness to the missing region

Table 7 shows all three models. Boosting loses the fewest points (11.3) mostly because it started low. Under masking, 41 of the 80 faces are predicted as identity 30. Boosting has a similar "sink" even on clean faces: identity 2 gets 9.4% of the total α on an average test face (four times its fair share), because weak trees often use it to label a big catch-all leaf, and 35 clean faces end up as 2. A zero block is darker than any real face, so many learners send masked faces the same way and identity 30 collects the votes.

Example: test face 17 (identity 1). Bagging is right on both versions (125 votes clean, 79 masked, best rival 19). Boosting is right when clean but says 30 when masked (q_1: $0.117 \to 0.077$, $q_{30} = 0.098$). *Bagging:* masking changes 46% of the trees' votes and disagreement on this face rises from 0.61 to 0.82, but the changed votes scatter over several people and identity 1 started far ahead, so it still wins. *Boosting:* learners holding 29% of the α change their vote, four of them away from identity 1 and none towards it. The real problem is the clean margin, only 0.01 over identity 2. Because the $(K-1)/K$ rule keeps moving the weight to whatever the last learner missed, each identity is backed by only a small slice of the α mass, and the learners also lean on central pixels slightly more than chance (8.1% of α-weighted splits in a block that is 6.25% of the image). A few flipped learners are enough. Bagging's large plurality absorbs the extra disagreement; boosting's thin, weight-driven margin does not.

4. Q3: Bias and variance as measured quantities

I keep the clean test set fixed and build 30 training sets, each drawing 6 faces with replacement from every person's 6 training faces (about 4 distinct faces per person). The three selected models are refitted on each set without retuning. The class vector is the leaf distribution (tree), the vote fractions (bagging) or q (boosting), and bias^2+variance is the mean squared distance to the one-hot target.

4.1. (a) Selected models

Table 8 gives the results. Accuracies are lower than for the frozen models because each set has fewer distinct images, but the ranking holds. The tree has by far the largest variance (0.67), and bagging removes 97% of it. Bagging's bias^2 is *higher* than the tree's (0.558 vs 0.475), which surprised me at first. The reason is that bias is measured on vectors: bagging's votes are spread out (winning share

	bias^2	variance	sum	test acc. (30 refits)
Tree	.475	.670	1.145	42.8 ± 5.0
Bagging	.558	.020	.578	71.7 ± 4.3
Boosting	.885	.018	.903	33.9 ± 7.3

Table 8. Measured bias^2 and variance on the clean test set, and mean $\pm$ standard deviation of test accuracy across the 30 refits.

about 0.36), so even its average vector is far from one-hot. Taking the argmax of the average vector instead, bagging is right on 85% of faces and the tree on 71%. The total error (0.58 vs 1.15) matches the accuracy ranking. Boosting has the lowest variance but the highest bias and the lowest, least stable accuracy (33.9 ± 7.3).

4.2. (b) Effect of depth

The selected tree is already unbounded, so for the "much deeper" tree I remove the leaf-size floor (leaf size 1). Depth 4 is the shallow tree, and I also ran depths 2, 8 and 16 for part (c) (Tab. 9, Fig. 7).

Going deeper halves the tree's bias^2 (0.93 to 0.48) while its variance grows about 16 times (0.04 to 0.67). Shallow trees have low variance only because, with 3–5 leaves, they predict almost the same thing every time.

Bagging removes 73% of the variance at depth 2, 83% at depth 4, 93% at depth 16 and 96% for unbounded trees (0.03, 0.06, 0.25 and 0.65 in absolute terms). Its own variance stays between 0.011 and 0.024 at every depth, and its bias^2 equals the tree's except for unbounded trees, for the reason given in (a). On these resampled sets the unbounded single tree is the same with leaf size 1 or 4, because with about four distinct faces per person each identity becomes one pure leaf. Inside bagging the floor does matter: leaf size 1 gives 13% singleton leaves and 3 points more accuracy.

4.3. (c) Is the deepest tree where bagging helps most?

No. The gain is largest at depth 16 (+34.6 points), smaller for unbounded trees (+27.8 and +30.8) and smallest for shallow ones (+10.4 at depth 2). With a standard error of about 1.1 points, the gap is real.

At depth 16 a tree has about 18 leaves of 13 images, so it can name only 18 people, and which 18 depends strongly on the sample. The bagged trees disagree a lot (0.894) and together cover everyone, so the removed variance (0.25) turns straight into accuracy. Unbounded trees lose even more variance (0.65), but they already have one leaf per person, agree more (0.834), and what is left looks like systematic confusion, such as identities 25 and 37 in Q1(d), which averaging cannot fix. Shallow trees disagree the most (0.95–0.96) but bias dominates (≥ 0.89): every tree misses most people, so there is little to average. The gain peaks where bias is moderate and disagreement is still high, which here is depth 16.

4.4. (d) Summary table

Table 10 brings the main numbers together.

Most stable: bagging. Its variance is a thirtieth of the tree's and its accuracy varies least across refits (± 4.3); the flat disagreement and margin curves in Fig. 3 say the same. Boosting's variance is slightly lower, but only because its q vectors are very flat (largest entry about 0.11). Its predictions actually swing the most (± 7.3, from 17.5% to 46.3%).

Most sensitive to the mask: the single tree. It loses 31.3 points, sends most masked faces to one leaf, and over the refits scores 23.5 ± 9.3 on masked faces. Bagging loses 13.8 points and stays the best masked model; boosting loses 11.3 but started low and falls into a sink class.

Is lowest variance best? No. Boosting has the lowest variance and the worst accuracy because its bias is the highest. Accuracy follows the total bias^2+variance (bagging 0.58, boosting 0.90, tree 1.15) much better than variance alone. Lower variance helps only when bias stays similar, which is exactly what bagging does for the tree.

References

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	single tree		bagging (100 trees)		var. removed	test accuracy (%)					
max_depth / leaf	bias^2	var	bias^2	var	abs. (frac.)	tree	bag	gain	disagr.	leaves	per leaf
2 / 4	.934	.041	.932	.011	.030 (73%)	5.1	15.5	+10.4	.961	3.0	80.0
4 / 4 (shallow)	.894	.073	.893	.012	.061 (83%)	8.8	28.5	+19.7	.949	5.2	46.2
8 / 4	.809	.146	.817	.016	.131 (89%)	15.0	43.3	+28.3	.928	9.5	25.4
16 / 4	.689	.271	.690	.020	.251 (93%)	25.2	59.8	**+34.6**	.894	17.9	13.4
None / 4 (selected)	.475	.670	.558	.024	.646 (96%)	42.8	70.5	+27.8	.834	40.0	6.0
None / 1 (deepest)	.475	.670	.537	.024	.646 (96%)	42.8	73.6	+30.8	.824	40.0	6.0

Table 9. Depth study over the 30 resampled training sets (means). "disagr." is the mean pairwise disagreement of the bagged trees on the test set; leaves and images per leaf are for the single tree. With `min_samples_leaf=1` the bagged trees have 46.5 leaves on average and 13.4% of their leaves hold one image; with 4 they have 39.2 leaves and none.

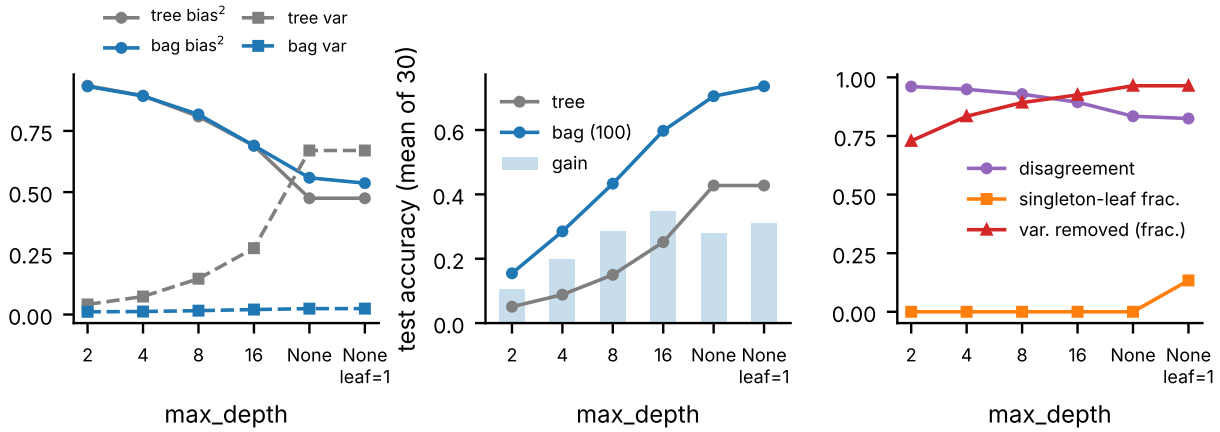


Figure 7. Depth study (means over 30 resampled training sets). Left: bias^2 (solid) and variance (dashed) for the single tree and 100-tree bagging. Middle: mean test accuracy and bagging's gain. Right: disagreement among the bagged trees, fraction of singleton leaves in the bagged trees, and fraction of the tree's variance removed by bagging.

	clean acc	masked acc	bias^2	variance
Tree (None, leaf 4)	55.0	23.8	.475	.670
Bagging (200)	**81.3**	**67.5**	.558	.020
Boosting (d4, 100)	46.3	35.0	.885	**.018**

Table 10. Frozen models: clean and masked test accuracy (%), with bias^2 and variance from the 30 refits. On the masked set the refit bias^2/variance are tree .794/.737, bagging .715/.021, boosting .895/.019.